

FULL STACK WEB DEVELOPMENT - SOURCE CODE

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Question 1: Interactive Product Calculator (HTML, CSS, JavaScript)

File: index.html

```
<!DOCTYPE html>
<!-- Program: Interactive Product Calculator
      Purpose: Calculate total price = quantity x price dynamically,
              without refreshing the page. -->
<html lang="en">
<head>
  <meta charset="UTF-8">
  <!-- Makes the page fit properly on mobile screens -->
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Interactive Product Calculator</title>
  <!-- CSS file -->
  <link rel="stylesheet" href="style.css">
</head>
<body>
  <div class="card">
    <h2>Product Calculator</h2>

    <!-- Input 1: product name (text) -->
    <label for="productName">Product Name</label>
    <input type="text" id="productName" value="Notebook">

    <!-- Input 2: quantity (whole number, minimum 1) -->
    <label for="quantity">Quantity</label>
    <input type="number" id="quantity" value="1" min="1" step="1">

    <!-- Input 3: price per unit (0 or more, two decimals allowed) -->
    <label for="price">Price per Unit (Rs.)</label>
    <input type="number" id="price" value="0" min="0" step="0.01">

    <!-- Error message area (empty when the input is valid) -->
    <div class="error" id="errorMessage"></div>

    <!-- Output area: summary line and total price -->
    <div class="result">
      <div class="summary" id="summary"></div>
      <div class="total" id="totalPrice">Total: Rs. 0.00</div>
    </div>
  </div>

  <!-- JavaScript file -->
  <script src="script.js"></script>
</body>
</html>
```

File: style.css

```
/* style.css - Interactive Product Calculator */

/* Page background and centering of the calculator card */
body {
  font-family: Arial, sans-serif;
  background-color: #eef2f7;
  display: flex;
  justify-content: center;
  padding: 40px 10px;
}
/* White card that holds the form */
.card {
  background: #ffffff;
  width: 380px;
  padding: 24px 28px;
  border-radius: 10px;
  box-shadow: 0 2px 10px rgba(0, 0, 0, 0.15);
}
h2 { text-align: center; color: #1f3a5f; margin-top: 0; }
label { display: block; margin-top: 14px; font-weight: bold; }
/* Input boxes take the full width of the card */
input {
  width: 100%;
  padding: 8px;
  margin-top: 5px;
  box-sizing: border-box;
  font-size: 15px;
}
/* Validation message is shown in red */
.error {
  color: #c62828;
  min-height: 20px;
  margin-top: 12px;
  font-size: 14px;
}
/* Green result box that displays the summary and total */
.result {
  margin-top: 10px;
  padding: 14px;
  background: #e8f5e9;
  border-left: 5px solid #2e7d32;
  border-radius: 4px;
}
.result .summary { font-size: 14px; color: #444; min-height: 18px; }
.result .total { font-size: 24px; font-weight: bold; color: #1b5e20; }
```

File: script.js

```
// script.js - Interactive Product Calculator

// Step 1: Get references to the HTML elements using the DOM
const productInput = document.getElementById("productName");
const quantityInput = document.getElementById("quantity");
const priceInput = document.getElementById("price");
const errorMessage = document.getElementById("errorMessage");
const summary = document.getElementById("summary");
const totalPrice = document.getElementById("totalPrice");

// Step 2: Function to validate the inputs and calculate the total price
function calculateTotal() {
    // Read the values; use "Product" if the name box is empty
    const productName = productInput.value.trim() || "Product";
    // Convert the text in the boxes to numbers
    const quantity = parseFloat(quantityInput.value);
    const price = parseFloat(priceInput.value);

    // Clear any old error message
    errorMessage.textContent = "";

    // Validation 1: quantity must be a whole number, minimum 1
    if (isNaN(quantity) || quantity < 1 || !Number.isInteger(quantity)) {
        errorMessage.textContent =
            "Quantity must be a whole number (minimum 1).";
        summary.textContent = "";
        totalPrice.textContent = "Total: Rs. 0.00";
        return; // stop here, do not calculate
    }
    // Validation 2: price must be a number and not negative
    if (isNaN(price) || price < 0) {
        errorMessage.textContent =
            "Price must be a valid number (0 or more).";
        summary.textContent = "";
        totalPrice.textContent = "Total: Rs. 0.00";
        return; // stop here, do not calculate
    }

    // Calculation: total = quantity x price
    const total = quantity * price;
    // Update the page (DOM) directly, so no page refresh is needed
    summary.textContent = quantity + " x " + productName +
        " @ Rs. " + price.toFixed(2);
    totalPrice.textContent = "Total: Rs. " + total.toFixed(2);
}

// Step 3: Event listeners - recalculate whenever the user changes
// any field ("input" fires on every key press or arrow click)
productInput.addEventListener("input", calculateTotal);
quantityInput.addEventListener("input", calculateTotal);
priceInput.addEventListener("input", calculateTotal);

// Step 4: Show the initial result when the page loads
calculateTotal();
```